

SAYAN JANA

AI/ML ENGINEER | MACHINE LEARNING ANALYST

6290316422 | sayanjanawd@gmail.com | **Linked In:** linkedin.com/in/sayanj1999 | **Portfolio:** sayanj-portfolio.netlify.app

PROFILE SUMMARY

Machine Learning and Data Analytics professional with a Master's degree in Data Science and hands-on experience in building AI-powered applications, recommendation systems, predictive analytics, and business intelligence solutions. Skilled in Python, Machine Learning, Natural Language Processing (NLP), Scikit-Learn, SQL, Power BI, and statistical analysis. Experienced in developing end-to-end data science solutions involving data preprocessing, feature engineering, model development, deployment, and visualization.

Built a content-based Movie Recommendation System leveraging NLP techniques, vectorization, cosine similarity, and TMDB API integration to deliver personalized movie suggestions through an interactive Streamlit application. Strong foundation in machine learning algorithms, data modeling, exploratory data analysis, and translating complex datasets into actionable insights. Passionate about developing scalable AI solutions that solve real-world business problems and enhance user experience.

EDUCATION

- **M.Sc. in Data Science** – Manipal Academy of Higher Education – 2023-2025 - 9.21 C.G.P.A.
- **B.Sc in Economics** – Calcutta University – 2018-2021
- **Higher Secondary Education** - CBSE – Apr, 2018
- **Secondary Education** - CBSE – Apr, 2016

EXPERIENCE

Business Analytics Intern – Labmentix (Sept 2025 – March 2026)

- Performed data cleaning, transformation, and validation to support business reporting and analytics workflows.
- Conducted exploratory data analysis (EDA) to identify customer trends, operational patterns, and performance insights.
- Built interactive Power BI dashboards for KPI monitoring and business performance tracking.
- Collaborated with stakeholders to translate business requirements into analytical reports and actionable insights.
- Applied statistical analysis and feature engineering techniques to improve dataset usability and reporting accuracy.
- Supported data-driven decision-making through visual storytelling and analytical presentations.

Web Designer – The Business Research Company (Oct 2022 – Sept 2023)

- Designed and developed responsive, user-friendly websites aligned with company branding using HTML, CSS, Bootstrap, and JavaScript.
- Improved website performance and usability through responsive design, optimized navigation, and page load efficiency improvements.

Graphics Designer – Alteus Biogenics Pvt. Ltd. (Nov 2020 – May 2022)

- Created a variety of design materials—including business cards, folders, banners, pamphlets, and social media creatives—while maintaining consistent, high-quality brand visuals.

TECHNICAL SKILLS

• Programming Languages

Python, SQL, R

• Data Science & Machine Learning

Machine Learning Fundamentals, Natural Language Processing (NLP), Recommendation Systems, Regression Analysis, Predictive Modeling, Feature Engineering, Statistical Modeling, Hypothesis Testing, Exploratory Data Analysis (EDA)

• Python Libraries

Pandas, NumPy, Scikit-Learn, Matplotlib, NLTK

• Business Intelligence & Visualization

Power BI, Tableau, DAX, Power Query, Advanced Excel, Dashboard Development, KPI Reporting, Data Visualization

• Database & Data Management

PostgreSQL, SQL Queries, Data Cleaning, Data Validation, Data Transformation, Data Modeling

• Professional Skills

Data Storytelling, Analytical Thinking, Problem Solving, Stakeholder Communication, Cross-Functional Collaboration

PROJECTS

Movie Recommendation System - Tools & Technologies: Python, Scikit-Learn, NLP, NLTK, Pandas, NumPy, Streamlit, TMDb API

- Developed a content-based recommendation engine using Natural Language Processing (NLP) techniques to suggest movies based on genres, keywords, cast, and crew information.
- Implemented text preprocessing, stemming, feature extraction, and vectorization using CountVectorizer and NLTK to create meaningful movie representations.
- Built a similarity model using Cosine Similarity to identify and recommend the most relevant movies from a dataset of 5000+ movies.
- Integrated TMDb API for dynamic retrieval and display of high-resolution movie posters, enhancing user experience and application interactivity.
- Designed and deployed an interactive Streamlit web application with custom UI components and efficient runtime similarity computation.
- Applied machine learning concepts including feature engineering, text mining, recommendation algorithms, and model optimization to deliver personalized recommendations.

Housing Price Prediction - Tools: Excel, Data Analysis ToolPak, Multiple Linear Regression, Statistical Analysis, Predictive Analytics, Data Visualization

- Built a multiple linear regression model in Excel to identify key drivers of housing market values.
- Performed statistical analysis, hypothesis testing, and coefficient interpretation on demographic, economic, and property-related variables.
- Achieved an R^2 of 0.52, explaining over 50% of housing value variation and providing insights for real estate valuation.

Ola Insights Analysis - Tools: PostgreSQL, Python, Pandas, Power BI, DAX, Streamlit

- Performed exploratory data analysis and feature engineering on ride-booking datasets.
- Built interactive dashboards to monitor revenue, ride success rates, customer behavior, and cancellation trends.
- Generated business insights through KPI tracking and visualization.

Medibuddy Insight Analysis - Tools: Python, SQL, Pandas, Matplotlib

- Conducted statistical analysis and exploratory data analysis on healthcare service datasets.
- Identified utilization patterns and operational trends through data-driven analysis.
- Developed visual reports supporting business decision-making.

IPL Performance Analytics Dashboard - Tools: Power BI, Power Query, DAX

- Built interactive sports analytics dashboards to evaluate player and team performance.
- Developed advanced DAX measures and performance metrics.
- Created data models and visualizations for trend analysis.

CERTIFICATION

- **Microsoft Power BI Data Analyst** – Microsoft (Coursera) – Oct, 2025
- **Databases and SQL for Data Science with Python** – IBM (Coursera) – Apr, 2025
- **Specialization: Business Statistics and Analysis** – Rice University (Coursera) – Apr, 2025
- **Introduction to Data Analysis Using Excel** – Rice University (Coursera) – Apr, 2025
- **Building Statistical Models in R: Linear Regression** – Coursera – May, 2024

LANGUAGES KNOWN

English | Bengali | Hindi